

Regularization Methods

Hyoungh Joon, Kim

June 3, 2026

Contents

1	Introduction	2
2	Main Ideas	2
2.1	Inductive Bias	2
2.2	Weight Decay	2
2.2.1	Generalized Weight Decay	2
2.2.2	Constant Regularizers	2
2.3	Early Stopping	3
2.3.1	Double Descent	3
2.4	Parameter Sharing	3
2.4.1	Soft Parameter Sharing	3
2.5	Dropout	3
3	The sections of this note may be created to individual documents later	4

1 Introduction

In this document, we discuss about methods of regularization. Regularization is used to deal with overfitting. There are several methods and these methods can be used together.

2 Main Ideas

2.1 Inductive Bias

Inductive bias is a preference for one choice over others. It is also called prior knowledge, literally. Inductive bias helps to constrain the space of solutions. To exploit the inductive bias, we use the following regularization approaches:

2.2 Weight Decay

Weight decay is used to control the size of the parameters. It prevents parameters being too large or too small. It is called weight decay, due to the following equation of **sum-of-squares regularizer**:

$$\nabla E(\mathbf{w}) + \lambda \mathbf{w}.$$

The sequential update will lead weights to zero.

2.2.1 Generalized Weight Decay

Definition 2.2.1 (Generalized Weight Decay)

A generalized weight decay regularizer is defined as follows:

$$\Omega(\mathbf{w}) = \frac{\lambda}{2} \sum_{j=1}^M |w_j|^q$$

$q = 2$ corresponds to the **L2 regularizer**, and $q = 1$ corresponds to the **L1 regularizer**. The regression with L2 regularizer is called **ridge regression**, and the one with L1 regularizer is called **lasso regression**. The lasso regression makes *sparse model*, since the lasso regression drives the coefficients or weights, to zero.

In the Bayesian perspective, ridge regression and lasso regression is same as setting the prior as Gaussian and Laplace with Gaussian likelihood, respectively.

Elastic net is regularizer which combines L2 and L1 regularizer;

$$\Omega(\mathbf{w}) = \frac{\lambda_1}{2} \sum_{j=1}^M |w_j| + \frac{\lambda_2}{2} \sum_{j=1}^M |w_j|^2$$

2.2.2 Constant Regularizers

The L2 regularizer and other regularizers, cannot include the inductive bias such as invariance and equivariance in itself. To deal with it, we must treat the weights and biases differently at each deep network layer. This method, which preserves

the invariance is called **constant regularizer**. But this method is not used so much.

2.3 Early Stopping

Early Stopping, is to stop the training at the point of smallest error with respect to validation error. Stopping a training before a minimum of the training error has been reached can be understood as limiting the effective network complexity.

2.3.1 Double Descent

However, there is a phenomenon called **double descent** (Belkin et al., 2019). The validation error decreases first, and increases a bit, then suddenly, it decreases again. This can be understood as the transition of regime;

1. Under-parameterized Regime: The complexity of the model is smaller than the number of samples. The model follows classical bias-variance tradeoff.
2. Over-parameterized Regime: The complexity of the model is large enough so that the test error is 0. In this case, the model's test error decreases as the model's complexity increases.

For more information, check Nakkiran et al., 2019. This paper tries to explain the double descent with **effective model complexity**, defined as following:

Definition 2.3.1 (Effective Model Complexity)

The **Effective Model Complexity** (EMC) of a training procedure $\mathcal{T}$, with respect to distribution $\mathcal{D}$ and parameter $\epsilon > 0$ is defined as:

$$\text{EMC}_{\mathcal{D},\epsilon}(\mathcal{T}) = \max \{n | \mathbb{E}_{S \sim \mathcal{D}^n} [\text{Error}_S(\mathcal{T}(S))] \leq \epsilon\}$$

where $\text{Error}_S(M)$ is the mean error of model M on train samples S .

(Cannot understand the meaning of it now...)

2.4 Parameter Sharing

Parameter Sharing is to group the parameters and let the grouped parameters to have the same value. Parameter sharing is used to encode some invariance or equivariance into a network. CNN is the example of the parameter sharing.

2.4.1 Soft Parameter Sharing

Instead of fixing the parameters to share same values, we can encourage the parameters in the same group to have similar values, leading to **soft parameter sharing**. We can use *mixture of Gaussians* to achieve this.

2.5 Dropout

Dropout is to delete the nodes from the network randomly during training. This method can be understood as implicit model averaging. Dropout can be applied to the input and hidden layers, but not the output layers. The probability

$p = 0.5$ of deleting the node, from the experiences, is known to work well with hidden layers. For the input layer, $p = 0.8$ is works well.

The dropout should be applied only when the model is being trained. However, there is a method called **Monte Carlo dropout**, where we still use dropout while inferencing. The Monte Carlo dropout simply make inferences mutiple times and average the results. Averaging 10 to 20 results is known to process good results. Monte Carlo dropout can provide us the uncertainty of the result, since we can check the standard deviation of the model's prediction.

3 The sections of this note may be created to individual documents later